

YEAR 7 MATHEMATICS

Chapter 2

Angles

WORKSHEETS

2

How to use this booklet

- 1** There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

WE DO **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

- 2** Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

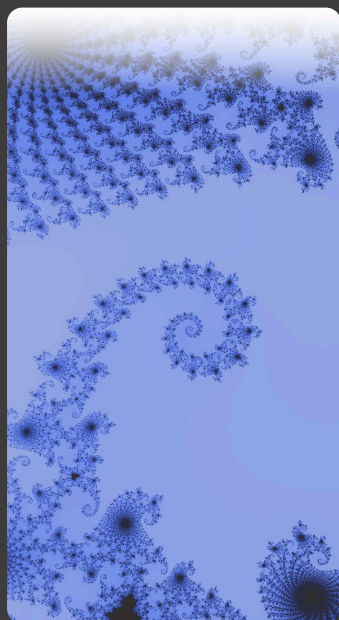
YOU DO **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
2.01	Naming angles	3		☐
2.02	Measuring and drawing angles	4		☐
2.03	Classifying angles	5		☐
2.04	Complementary and supplementary angles	6		☐
2.05	Angles at a point and vertically opposite angles	7		☐
2.06	Parallel and perpendicular lines	8		☐
2.07	Corresponding angles	9		☐
2.08	Alternate angles	10		☐
2.09	Co-interior angles	11		☐
2.10	Finding unknown angles on parallel lines	12		☐
2.11	Proving lines are parallel	13		☐

In this chapter you will

- measure and draw angles with a protractor
- classify angles by size
- find complementary, supplementary and vertically opposite angles
- construct parallel and perpendicular lines
- find corresponding, alternate and co-interior angles on parallel lines
- prove that two lines are parallel



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is Seahorse Valley, the curling tails between the main body and the round bulb on its left. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

Easy

Questions 1–15

Which letter is the vertex? It is always the middle letter.

WE DO With your teacher: copy the working.

E1 $\angle PEC$

E2 $\angle PAQ$

YOU DO Your turn.

1 $\angle GYW$

2 $\angle RMZ$

3 $\angle AWC$

4 $\angle EMC$

5 $\angle AXZ$

6 $\angle EXB$

7 $\angle ZUE$

8 $\angle XSW$

9 $\angle UBS$

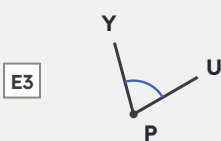
10 $\angle FZQ$

11 $\angle KHR$

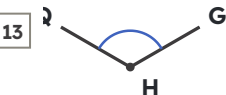
12 $\angle SKU$

Name the marked angle with three letters. Put the vertex in the middle.

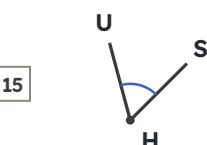
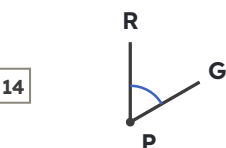
WE DO With your teacher: copy the working.



YOU DO Your turn.



$\angle GHQ$ or $\angle QHG$



Medium

Questions 16–25

Name each angle another way, then write its vertex and its two arms.

WE DO With your teacher: copy the working.

E1 $\angle HBX$

E2 $\angle FYW$

YOU DO Your turn.

16 $\angle ZRC$

also $\angle CRZ$

vertex R, arms RZ and RC

17 $\angle YVQ$

also $\angle QVY$

18 $\angle NLU$

19 $\angle WUL$

20 $\angle SLC$

21 $\angle VGH$

22 $\angle MTS$

23 $\angle EBX$

24 $\angle EAH$

25 $\angle GAW$

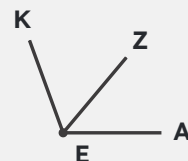
Challenging

Questions 26–29

Use the diagram. Name the angles and the arms.

WE DO With your teacher: copy the working.

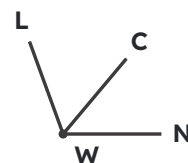
E1 Name the three angles at E.



E2 $\angle PME$ and $\angle EMY$ are next to each other. Name the vertex and the arm they share.

YOU DO Your turn.

26 Name the three angles at W.



small angles: $\angle NWC$, $\angle CWL$

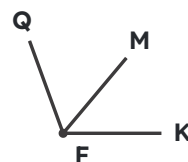
whole angle: _____

27 $\angle YMQ$ and $\angle QMP$ are next to each other. Name the vertex and the arm they share.

vertex: _____

common arm: _____

28 Name the three angles at F.



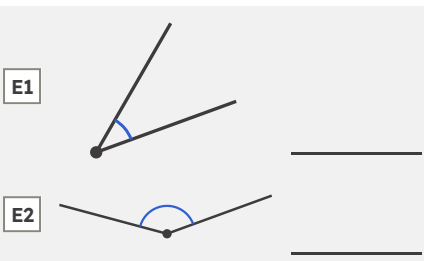
29 $\angle HUR$ and $\angle RUT$ are next to each other. Name the vertex and the arm they share.

Easy

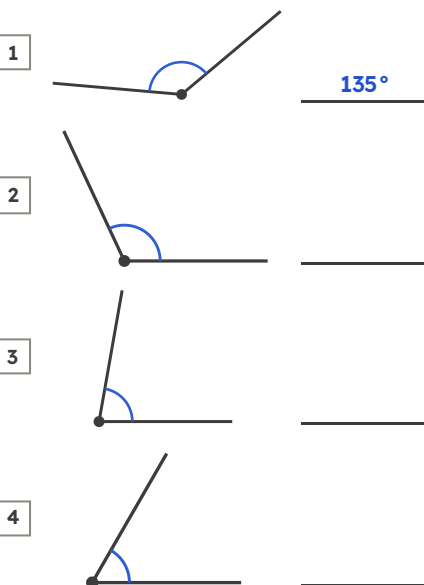
Questions 1–15

Measure each angle with your protractor. Line up the base line on one arm.

WE DO With your teacher: copy the working.



YOU DO Your turn.



To draw a reflex angle, draw 360° – the angle. Write the angle you draw.

WE DO With your teacher: copy the working.

E3 285° _____

YOU DO Your turn.

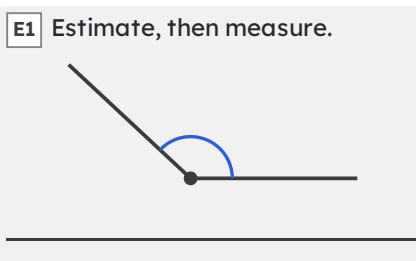
- | | | |
|----|-------------|-------|
| 5 | 355° | _____ |
| 6 | 325° | _____ |
| 7 | 250° | _____ |
| 8 | 240° | _____ |
| 9 | 190° | _____ |
| 10 | 210° | _____ |
| 11 | 200° | _____ |
| 12 | 235° | _____ |
| 13 | 205° | _____ |
| 14 | 275° | _____ |
| 15 | 230° | _____ |

Medium

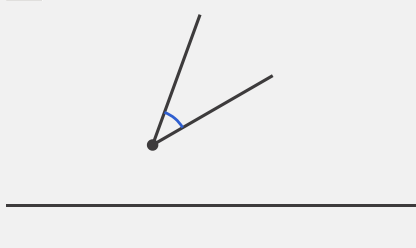
Questions 16–18

Estimate the angle first, then measure it with your protractor.

WE DO With your teacher: copy the working.

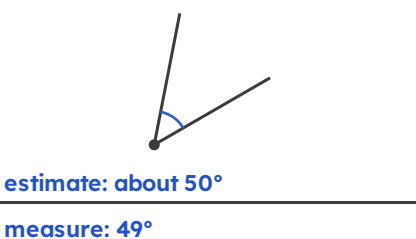


E2 Estimate, then measure.

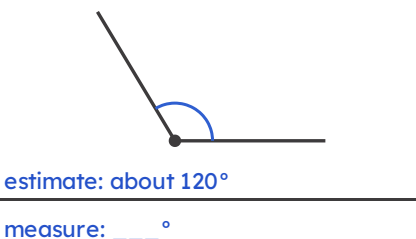


YOU DO Your turn.

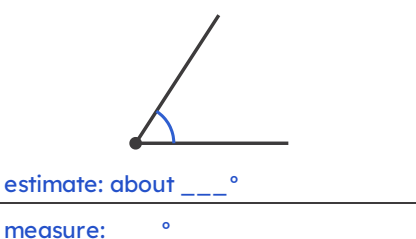
16 Estimate, then measure.



17 Estimate, then measure.



18 Estimate, then measure.



Challenging

Questions 19–25

Plan each drawing in steps. Then draw it on the back of the sheet.

WE DO With your teacher: copy the working.

E1 How do you draw a reflex angle of 200° ?

E2 An angle is 60° less than a revolution. Find its size and classify it.

YOU DO Your turn.

19 How do you draw a reflex angle of 225° ?

$$360^\circ - 225^\circ = 135^\circ$$

draw 135° , then mark the reflex angle outside

20 An angle is 40° less than a revolution. Find its size and classify it.

$$360^\circ - 40^\circ = 320^\circ$$

320° is reflex

21 How do you draw a reflex angle of 310° ?

22 An angle is 75° less than a revolution. Find its size and classify it.

23 How do you draw a reflex angle of 325° ?

24 An angle is 70° less than a revolution. Find its size and classify it.

25 How do you draw a reflex angle of 240° ?

Easy

Questions 1–16

Classify each angle: acute, right, obtuse, straight, reflex or revolution.

WE DO With your teacher: copy the working.

E1 360°

E2 90°

YOU DO Your turn.

1 20°

acute

2 180°

3 95°

4 340°

5 130°

6 270°

7 110°

8 55°

9 235°

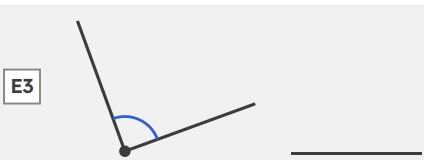
10 185°

11 140°

12 295°

Classify the angle drawn.

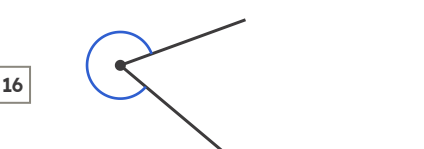
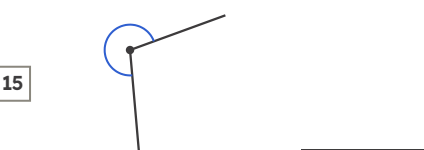
WE DO With your teacher: copy the working.



YOU DO Your turn.



right



Medium

Questions 17–38

Add the two angles, then classify the total.

WE DO With your teacher: copy the working.

E1 $155^\circ + 95^\circ$

E2 $150^\circ + 160^\circ$

YOU DO Your turn.

17 $120^\circ + 50^\circ$

18 $180^\circ + 150^\circ$

$120^\circ + 50^\circ = 170^\circ$

$180^\circ + 150^\circ = 330^\circ$

170° is obtuse

330° is reflex

19 $55^\circ + 95^\circ$

20 $75^\circ + 75^\circ$

$55^\circ + 95^\circ = 150^\circ$

150° is obtuse

21 $175^\circ + 10^\circ$

22 $25^\circ + 95^\circ$

23 $105^\circ + 50^\circ$

24 $70^\circ + 125^\circ$

25 $160^\circ + 160^\circ$

26 $90^\circ + 105^\circ$

27 $95^\circ + 40^\circ$

28 $125^\circ + 25^\circ$

29 $75^\circ + 50^\circ$

30 $20^\circ + 15^\circ$

31 $100^\circ + 15^\circ$

32 $30^\circ + 145^\circ$

33 $185^\circ + 30^\circ$

34 $130^\circ + 30^\circ$

35 $90^\circ + 40^\circ$

36 $70^\circ + 90^\circ$

37 $100^\circ + 40^\circ$

38 $150^\circ + 10^\circ$

Challenging

Questions 39–45

Work out the size of each angle, then classify it.

WE DO With your teacher: copy the working.

E1 The angle between the hands of a clock at 10:00.

E2 A right angle plus 20° .

E3 A straight angle plus 10° .

YOU DO Your turn.

39 The angle between the hands of a clock at 9:00.

each hour mark: $360^\circ \div 12 = 30^\circ$

$3 \times 30^\circ = 90^\circ$

40 A right angle plus 45° .

$90^\circ + 45^\circ = 135^\circ$

41 A straight angle plus 50° .

42 The angle between the hands of a clock at 8:00.

43 A right angle plus 10° .

44 A straight angle plus 30° .

45 The angle between the hands of a clock at 7:00.

Easy

Questions 1–24

Find the complement.
Complementary angles add to 90° .

WE DO With your teacher: copy the working.

E1 75° _____

E2 30° _____

YOU DO Your turn.

1 13° _____ 77°

2 72° _____

3 55° _____

4 70° _____

5 81° _____

6 67° _____

7 68° _____

8 25° _____

9 5° _____

10 62° _____

11 16° _____

12 85° _____

Find the supplement.
Supplementary angles add to 180° .

WE DO With your teacher: copy the working.

E3 140° _____

YOU DO Your turn.

13 105° _____ 75°

14 69° _____

15 75° _____

16 136° _____

17 27° _____

18 50° _____

19 150° _____

20 74° _____

21 173° _____

22 56° _____

23 178° _____

24 19° _____

Medium

Questions 25–33

Write an equation, then find the value of the pronumeral.

WE DO With your teacher: copy the working.

E1 p° and 75° are complementary.

E2 p° and 17° are supplementary.

YOU DO Your turn.

25 x° and 45° are complementary.

$x + 45 = 90$

$x = 90 - 45 = 45$

26 m° and 28° are supplementary.

$m + 28 = 180$

$m = ______ - ______ = ______$

27 p° and 16° are complementary.

$p + ______ = ______$

$p = ______ - ______ = ______$

28 m° and 45° are supplementary.

29 p° and 36° are complementary.

30 y° and 40° are supplementary.

31 x° and 75° are complementary.

32 p° and 33° are supplementary.

33 p° and 40° are complementary.

Challenging

Questions 34–35

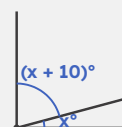
Write an equation from the diagram, then solve it.

WE DO With your teacher: copy the working.

E1 Find x .



E2 Find x .



YOU DO Your turn.

34 Find x .

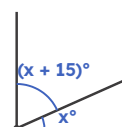


$3x + 60 = 180$

$3x = 120$

$x = ______$

35 Find x .



$x + x + ______ = ______$

$______ x = ______$

$x = ______$

Easy

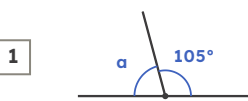
Questions 1–7

Angles on a straight line add to 180° . Find a .

WE DO With your teacher: copy the working.



YOU DO Your turn.

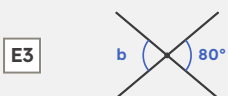


$$a = 75$$

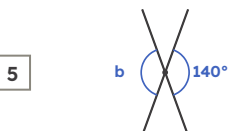


Vertically opposite angles are equal. Find b .

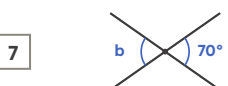
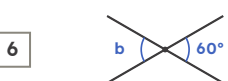
WE DO With your teacher: copy the working.



YOU DO Your turn.



$$b = 140$$



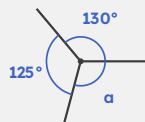
Medium

Questions 8–10

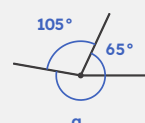
Angles at a point add to 360° . Write an equation, then find a .

WE DO With your teacher: copy the working.

E1 Find a .

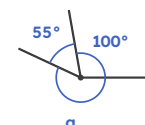


E2 Find a .



YOU DO Your turn.

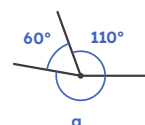
8 Find a .



$$a + 100 + 55 = 360$$

$$a = 360 - 155 = 205$$

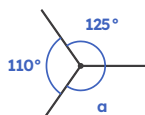
9 Find a .



$$a + 110 + 60 = 360$$

$$a = \underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

10 Find a .



$$a + \underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

$$a = \underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

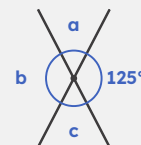
Challenging

Questions 11–12

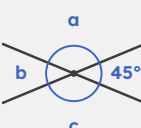
Find each pronumeral. Give a reason for each one.

WE DO With your teacher: copy the working.

E1 Find a , b and c .

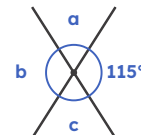


E2 Find a , b and c .



YOU DO Your turn.

11 Find a , b and c .

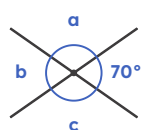


$$b = 115 \text{ (vertically opposite)}$$

$$a = 180 - 115 = 65 \text{ (straight line)}$$

$$c = \underline{\hspace{1cm}} \text{ (vertically opposite } a)$$

12 Find a , b and c .



$$b = \underline{\hspace{1cm}} \text{ (vertically opposite)}$$

$$a = \underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ (straight line)}$$

$$c = \underline{\hspace{1cm}} \text{ (vertically opposite } a)$$

Easy

Questions 1–22

Write the statement with a symbol: $\parallel$ means parallel, $\perp$ means perpendicular.

WE DO With your teacher: copy the working.

E1 KL is parallel to MN _____

E2 XY is perpendicular to ZW _____

YOU DO Your turn.

1 EF is parallel to GH EF $\parallel$ GH

2 AB is perpendicular to CD _____

3 PQ is parallel to RS _____

4 EF is perpendicular to GH _____

5 PQ is perpendicular to RS _____

6 XY is parallel to ZW _____

7 TU is parallel to VW _____

8 TU is perpendicular to VW _____

9 AB is parallel to CD _____

10 KL is perpendicular to MN _____

Are the lines in each object parallel ($\parallel$), perpendicular ($\perp$) or neither?

WE DO With your teacher: copy the working.

E3 train tracks _____

YOU DO Your turn.

11 the letter T $\perp$

12 the letter X _____

13 an equals sign _____

14 a plus sign _____

15 the letter V _____

16 the letter H (sides) _____

17 the letter L _____

18 a wall and floor _____

19 lines on ruled paper _____

20 the letter K _____

21 the edges of a ruler _____

22 the letter Z (top and bottom) _____

Medium

Questions 23–25

Use the shape. Name the pairs of parallel sides, then the pairs of perpendicular sides (write none if there are none).

WE DO With your teacher: copy the working.

E1 Rectangle TUVW

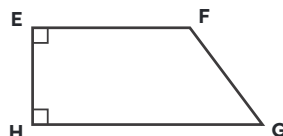


E2 Parallelogram KLMN



YOU DO Your turn.

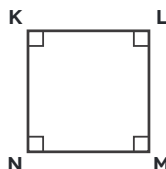
23 Right trapezium EFGH



$\parallel$: EF $\parallel$ HG

$\perp$: EH $\perp$ EF, EH $\perp$ HG

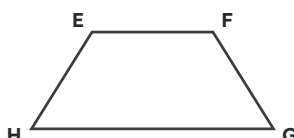
24 Square KLMN



$\parallel$: KL $\parallel$ NM, KN $\parallel$ LM

$\perp$: _____

25 Trapezium EFGH



$\parallel$: _____

$\perp$: _____

Challenging

Questions 26–29

Write the steps for each construction. Use a ruler and compasses, then draw it on the back of the sheet.

WE DO With your teacher: copy the working.

E1 Construct a line perpendicular to PQ through the point G on PQ.

E2 Construct a line parallel to PQ through the point H (not on PQ).

E3 Construct the perpendicular bisector of the interval CD.

YOU DO Your turn.

26 Construct a line perpendicular to CD through the point W on CD.

compasses at W: mark two points on CD

from each point: equal arcs that cross

27 Construct a line parallel to PQ through the point S (not on PQ).

rule the new arm: _____

28 Construct the perpendicular bisector of the interval EF.

29 Construct a line perpendicular to PQ through the point T on PQ.

Easy

Questions 1–12

The lines are parallel. Corresponding angles are equal. Write the corresponding angle.

WE DO With your teacher: copy the working.

E1 corresponding to 76° _____

E2 corresponding to 138° _____

YOU DO Your turn.

1 corresponding to 149° 149°

2 corresponding to 113° _____

3 corresponding to 39° _____

4 corresponding to 51° _____

5 corresponding to 28° _____

6 corresponding to 43° _____

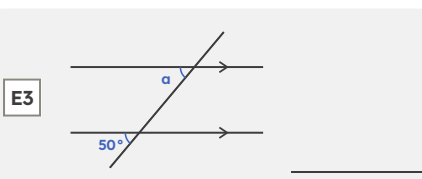
7 corresponding to 118° _____

8 corresponding to 109° _____

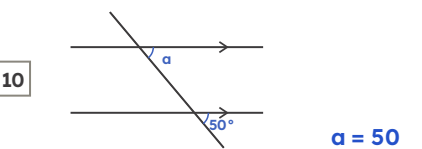
9 corresponding to 65° _____

The lines are parallel. Find a . (Look for the F shape.)

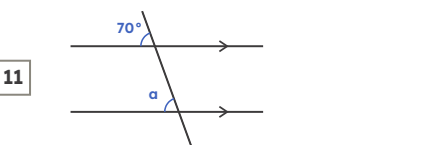
WE DO With your teacher: copy the working.



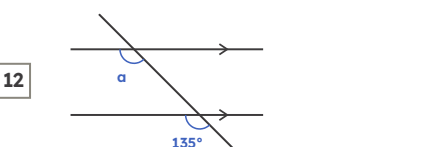
YOU DO Your turn.



10 $a = 50$



11 _____



12 _____

Medium

Questions 13–22

Find a and give the reason. Then find b , the angle next to a on the line.

WE DO With your teacher: copy the working.

E1 a is corresponding to 135° .

E2 a is corresponding to 100° .

YOU DO Your turn.

13 a is corresponding to 115° .

$a = 115$ (corresponding angles, parallel lines)

$b = 180 - 115 = 65$ (straight line)

14 a is corresponding to 145° .

$a = 145$ (corresponding angles, parallel lines)

$b = ______ - ______ = ______$ (straight line)

15 a is corresponding to 110° .

$a = ______$ (corresponding angles, parallel lines)

$b = ______ - ______ = ______$ (straight line)

16 a is corresponding to 80° .

17 a is corresponding to 95° .

18 a is corresponding to 125° .

19 a is corresponding to 150° .

20 a is corresponding to 70° .

21 a is corresponding to 85° .

22 a is corresponding to 50° .

Challenging

Questions 23–28

Corresponding angles are equal. Write an equation, then solve it.

WE DO With your teacher: copy the working.

E1 Corresponding angles are $(4x + 10)^\circ$ and 114° . Find x .

E2 Corresponding angles are $(2x + 15)^\circ$ and 65° . Find x .

YOU DO Your turn.

23 Corresponding angles are $(3x + 30)^\circ$ and 45° . Find x .

$$3x + 30 = 45$$

$$3x = 15$$

$$x = ______$$

24 Corresponding angles are $(3x + 30)^\circ$ and 75° . Find x .

$$______x + ______ = ______$$

$$______x = ______$$

$$x = ______$$

25 Corresponding angles are $(3x + 25)^\circ$ and 49° . Find x .

26 Corresponding angles are $(3x + 20)^\circ$ and 68° . Find x .

27 Corresponding angles are $(3x + 10)^\circ$ and 94° . Find x .

28 Corresponding angles are $(3x + 20)^\circ$ and 110° . Find x .

Easy

Questions 1–15

The lines are parallel.
Alternate angles are equal.
Write the alternate angle.

WE DO With your teacher: copy the working.

E1 alternate to 65° _____

E2 alternate to 33° _____

YOU DO Your turn.

1 alternate to 127° 127°

2 alternate to 108° _____

3 alternate to 82° _____

4 alternate to 159° _____

5 alternate to 154° _____

6 alternate to 48° _____

7 alternate to 66° _____

8 alternate to 24° _____

9 alternate to 133° _____

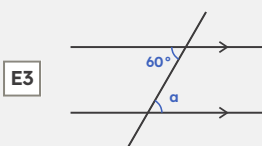
10 alternate to 156° _____

11 alternate to 106° _____

12 alternate to 36° _____

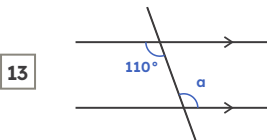
The lines are parallel. Find a .
(Look for the Z shape.)

WE DO With your teacher: copy the working.



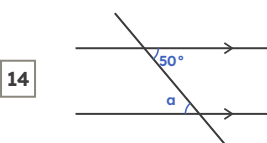
E3

YOU DO Your turn.

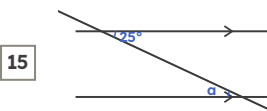


13

$a = 110$



14



15

Medium

Questions 16–25

Find a and give the reason.
Then find b , the angle next to a on the line.

WE DO With your teacher: copy the working.

E1 a is alternate to 120° .

E2 a is alternate to 70° .

YOU DO Your turn.

16 a is alternate to 155° .

$a = 155$ (alternate angles, parallel lines)

$b = 180 - 155 = 25$ (straight line)

17 a is alternate to 80° .

$a = 80$ (alternate angles, parallel lines)

$b = _____ - _____ = _____$ (straight line)

18 a is alternate to 105° .

$a = _____$ (alternate angles, parallel lines)

$b = _____ - _____ = _____$ (straight line)

19 a is alternate to 100° .

20 a is alternate to 140° .

21 a is alternate to 55° .

22 a is alternate to 125° .

23 a is alternate to 40° .

24 a is alternate to 45° .

25 a is alternate to 30° .

Challenging

Questions 26–31

Alternate angles are equal.
Write an equation, then solve it.

WE DO With your teacher: copy the working.

E1 Alternate angles are $(4x - 15)^\circ$ and 61° . Find x .

E2 Alternate angles are $(3x - 25)^\circ$ and 74° . Find x .

YOU DO Your turn.

26 Alternate angles are $(4x - 10)^\circ$ and 26° . Find x .

$4x - 10 = 26$

$4x = 36$

$x = _____$

27 Alternate angles are $(2x - 20)^\circ$ and 32° . Find x .

$_____x - _____ = _____$

$_____x = _____$

$x = _____$

28 Alternate angles are $(4x - 5)^\circ$ and 67° . Find x .

29 Alternate angles are $(2x - 15)^\circ$ and 49° . Find x .

30 Alternate angles are $(3x - 20)^\circ$ and 70° . Find x .

31 Alternate angles are $(2x - 15)^\circ$ and 19° . Find x .

Easy

Questions 1–15

The lines are parallel. Co-interior angles add to 180° . Write the co-interior angle.

WE DO With your teacher: copy the working.

E1 co-interior to 59° _____

E2 co-interior to 128° _____

YOU DO Your turn.

1 co-interior to 157° 23°

2 co-interior to 144° _____

3 co-interior to 46° _____

4 co-interior to 55° _____

5 co-interior to 142° _____

6 co-interior to 41° _____

7 co-interior to 31° _____

8 co-interior to 132° _____

9 co-interior to 114° _____

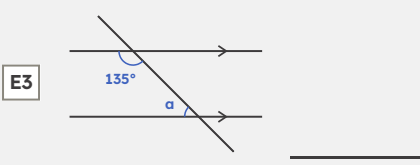
10 co-interior to 140° _____

11 co-interior to 103° _____

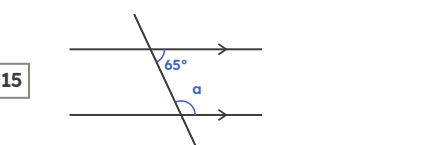
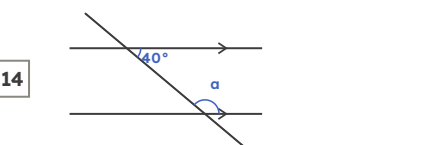
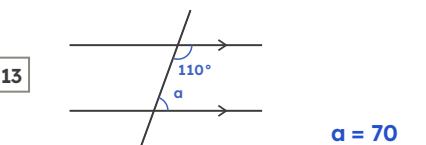
12 co-interior to 119° _____

The lines are parallel. Find a . (Look for the C shape.)

WE DO With your teacher: copy the working.



YOU DO Your turn.



Medium

Questions 16–25

Write an equation, then find a . Give the reason.

WE DO With your teacher: copy the working.

E1 a is co-interior to 38° .

E2 a is co-interior to 94° .

YOU DO Your turn.

16 a is co-interior to 69° .

$$a + 69 = 180 \text{ (co-interior angles)}$$

$$a = 180 - 69 = 111$$

17 a is co-interior to 57° .

$$a + 57 = 180 \text{ (co-interior angles)}$$

$$a = ___ - ___ = ___$$

18 a is co-interior to 103° .

$$a + ___ = ___ \text{ (co-interior angles)}$$

$$a = ___ - ___ = ___$$

19 a is co-interior to 30° .

20 a is co-interior to 27° .

21 a is co-interior to 77° .

22 a is co-interior to 136° .

23 a is co-interior to 117° .

24 a is co-interior to 45° .

25 a is co-interior to 71° .

Challenging

Questions 26–26

Co-interior angles add to 180° . Write an equation, then solve it.

WE DO With your teacher: copy the working.

E1 Co-interior angles are y° and $2y^\circ$. Find y and both angles.

E2 Co-interior angles are y° and $3y^\circ$. Find y and both angles.

YOU DO Your turn.

26 Co-interior angles are y° and $4y^\circ$. Find y and both angles.

$$y + 4y = 180$$

$$5y = 180, \text{ so } y = 36$$

$$\text{angles: } ___^\circ \text{ and } ___^\circ$$

2.10 Finding unknown angles on parallel lines

YEAR 7 · CHAPTER 2: ANGLES

Easy

Questions 1–11

Which rule? Write corresponding, alternate or co-interior.

WE DO With your teacher: copy the working.

E1 F shape

E2 Z shape

YOU DO Your turn.

1 C shape

co-interior

2 same position at each crossing

3 between the lines, opposite sides

4 between the lines, same side

5 the angles add to 180°

6 backwards Z (N shape)

7 upside-down F

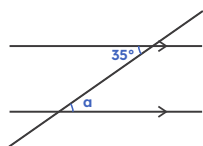
The lines are parallel. Find a .

WE DO With your teacher: copy the working.

E3

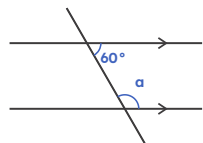
YOU DO Your turn.

8

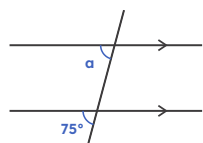


$$a = 35$$

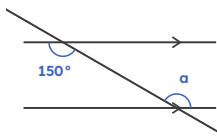
9



10



11



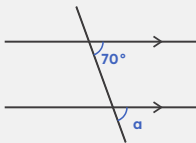
Medium

Questions 12–13

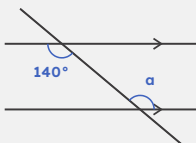
The lines are parallel. Find a , then write the rule you used.

WE DO With your teacher: copy the working.

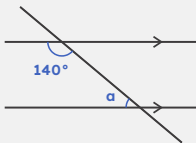
E1 Find a .



E2 Find a .

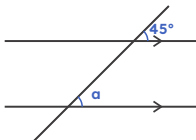


E3 Find a .



YOU DO Your turn.

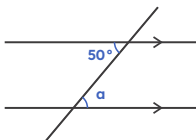
12 Find a .



$$a = 45$$

corresponding angles are equal

13 Find a .



$$a = 50$$

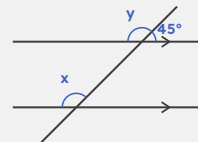
Challenging

Questions 14–16

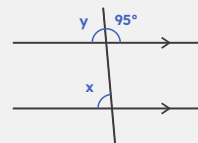
Find each pronumeral in two steps. Give a reason for each step.

WE DO With your teacher: copy the working.

E1 Find y , then x .

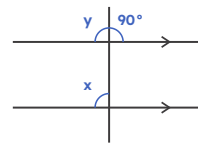


E2 Find y , then x .



YOU DO Your turn.

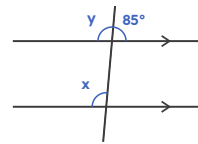
14 Find y , then x .



$$y = 180 - 90 = 90 \text{ (straight line)}$$

$$x = \text{---} \text{ (corresponding to } y \text{)}$$

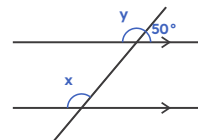
15 Find y , then x .



$$y = \text{---} - \text{---} = \text{---} \text{ (straight line)}$$

$$x = \text{---} \text{ (corresponding to } y \text{)}$$

16 Find y , then x .



2.11 Proving lines are parallel

YEAR 7 · CHAPTER 2: ANGLES

Easy

Questions 1–18

Is $AB \parallel CD$? Write yes or no. Equal corresponding or alternate angles, or co-interior angles adding to 180° , mean yes.

WE DO With your teacher: copy the working.

E1 alternate: 75° and 105°

E2 co-interior: 80° and 100°

YOU DO Your turn.

1 alternate: 65° and 65°

yes

2 alternate: 35° and 40°

3 co-interior: 85° and 95°

4 alternate: 35° and 35°

5 co-interior: 70° and 110°

6 alternate: 70° and 110°

7 alternate: 135° and 135°

8 corresponding: 30° and 150°

9 corresponding: 100° and 100°

10 alternate: 60° and 120°

11 co-interior: 85° and 90°

12 alternate: 55° and 55°

13 corresponding: 125° and 130°

14 co-interior: 140° and 40°

15 corresponding: 130° and 130°

16 co-interior: 40° and 140°

17 alternate: 70° and 75°

18 corresponding: 30° and 30°

Medium

Questions 19–26

Is $AB \parallel CD$? Check the angle pair, then give the reason.

WE DO With your teacher: copy the working.

E1 Corresponding angles: 60° and 65°

E2 Alternate angles: 75° and 80°

E3 Co-interior angles: 130° and 50°

YOU DO Your turn.

19 Corresponding angles: 65° and 65°

$$65 = 65$$

yes: corresponding angles are equal

20 Alternate angles: 35° and 35°

$$35 = 35$$

yes: _____

21 Co-interior angles: 150° and 30°

$$150 + 30 = 180$$

yes: co-interior angles add to 180°

22 Corresponding angles: 85° and 85°

23 Alternate angles: 85° and 85°

24 Co-interior angles: 70° and 110°

25 Corresponding angles: 130° and 135°

26 Alternate angles: 30° and 30°

Challenging

Questions 27–32

Find the value of x that makes $AB \parallel CD$. Show the equation.

WE DO With your teacher: copy the working.

E1 Alternate angles are $(3x + 10)^\circ$ and 97° .

E2 Co-interior angles are $2x^\circ$ and 160° .

YOU DO Your turn.

27 Alternate angles are $(3x + 10)^\circ$ and 61° .

$$3x + 10 = 61$$

$$3x = 51$$

$$x = 17$$

28 Co-interior angles are $2x^\circ$ and 150° .

$$2x + 150 = 180$$

$$2x = 30$$

$$x = 15$$

29 Alternate angles are $(3x + 10)^\circ$ and 28° .

30 Co-interior angles are $2x^\circ$ and 146° .

31 Alternate angles are $(3x + 10)^\circ$ and 34° .

32 Co-interior angles are $2x^\circ$ and 156° .